

DEMAND, SUPPLY & MARKET EQUILIBRIUM

Significance:

- The tools of **demand** and **supply** can be applied to a range of important topics such as:
 - Evaluating how **global weather conditions** will affect agricultural production and market prices of agricultural commodities;
 - Assessing the impact of government rent control on the **housing market**;

Cont.

- Understanding how taxes, subsidies, and other government policies affect both consumers and producers.
- Demand and supply analysis deals with how prices of products and resources are determined.

The Concept of DEMAND:

- **Demand** - refers to the various quantities of a good or service that consumers are **willing** and **able** to purchase at alternative prices, *ceteris paribus* (other things remaining equal).
 - It conveys both the elements of **desire** for the commodity and **capacity** to pay (must be willing and able).
 - It emphasizes the relationship between quantity bought and its price, although there may be other factors that determine how much a consumer wants to purchase.

The Law of Demand:

- Asserts that the **quantity demanded** of a good or service is **negatively** or **inversely** related to its **own price** – the First Law of Demand.
 - When the price increases, less of the good or service will be bought.
 - When the price decreases, more of the commodity will be purchased.

WHY SO?

Two Reasons for the Inverse Relationship:

■ Substitution effect

- When price of a good decreases, the consumer substitutes the lower priced good for the more expensive ones.

■ Income effect

- When price decreases, the consumer's real income (or purchasing power) increases, so he tends to buy more of the good.

P ↓ Q ↑

Presentation of the Demand Relationship:

The relationship between *quantity purchased* and alternative *prices* may be presented in 3 ways:

- Demand schedule – in tabular form.
- Demand curve – in graphical form.
- Demand function – in equation form.

Demand Schedule:

Demand Schedule for Shoes	
Price of a pair of shoes (in pounds)	Quantity demanded/year
0	8
50	7
100	6
150	5
200	4
250	3
300	2
350	1
400	0

Demand Curve:

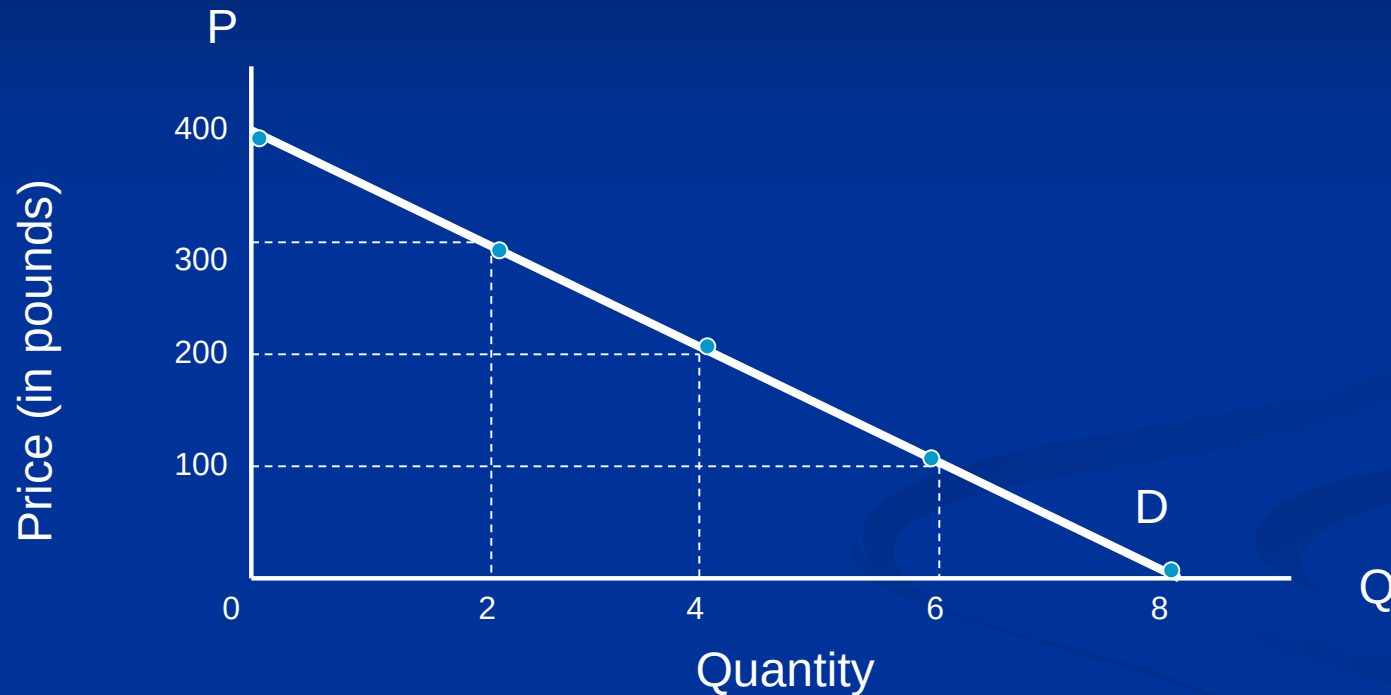


Figure 1: Demand Curve. The negative slope of the demand curve depicts the inverse relationship between price and quantity demanded.

Demand Function:

- Quantity demanded (Q) is expressed as a mathematical function of price (P). The demand function may thus be written as:

$$Q_d = a - bP$$

where:-

- a is the horizontal intercept of the equation or the quantity demanded when price is zero.
- $(-b)$ is the slope of the function.
- Example: $Q_d = 8 - 0.02P$

Important Factors Affecting Demand:

1. Price of the commodity.
2. Consumer incomes.
3. Prices of related commodities (substitutes and complements).
4. Tastes and preferences.
5. Consumer expectations.
6. Number of consumers.

(Cont.)

■ **Income:**

As income changes, demand for a commodity usually changes.

- Normal goods – are goods whose demand responds positively to changes in income.
 - Most goods are normal goods. As income increases, more of shoes, TVs, clothes, are bought.
- Inferior goods – are goods whose demand responds negatively to changes in income.
 - Few but existent. Examples: old cell phone models, used cars, some food items.

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■ Prices of Related Commodities in Consumption:

- Substitutes – are goods that are substitutable with each other (not necessarily perfect).
 - Examples are coffee and tea, Coke and Pepsi.
 - When the price of a substitute increases (P_y), quantity bought of the other good (Q_x) increases - (direct relationship)
- Complements – are goods that are used or consumed together.
 - Examples are coffee and sugar, bread and butter, tennis rackets and tennis balls, computers and software packages.
 - When the price of a complement increases, quantity bought of the other good decreases - (inverse relationship).

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■ Consumer Tastes and Preferences:

- When consumer tastes **shift towards** a particular good, greater amounts of a good are demanded at each price.
 - Example: if consumers' preference for drinking bottled water increases its demand curve will shift rightward.
- If consumer preferences change **away from a good**, its demand will decrease. At every possible price less of the good is demanded than before.
 - Example: the demand for cassette tapes decreased due to preference for DVDs.

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■ Consumer Expectations:

Expectations about **future prices** and **incomes** affect current demand for many goods and services.

- If we expect price of sugar to **increase**, we might stock up on the good to avoid the expected price increase. Thus, current demand for sugar might increase.
- Those who expect to **lose their jobs** due to bad economic conditions, will reduce their demand for a variety of goods in the current period.

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■ Number of Consumers:

It affects the total demand for a good.

- **Total demand** is also known as **market demand**. It is the **horizontal summation** of the individual demands of all consumers.

- An increase in the number of consumers shifts the market demand curve to the right.

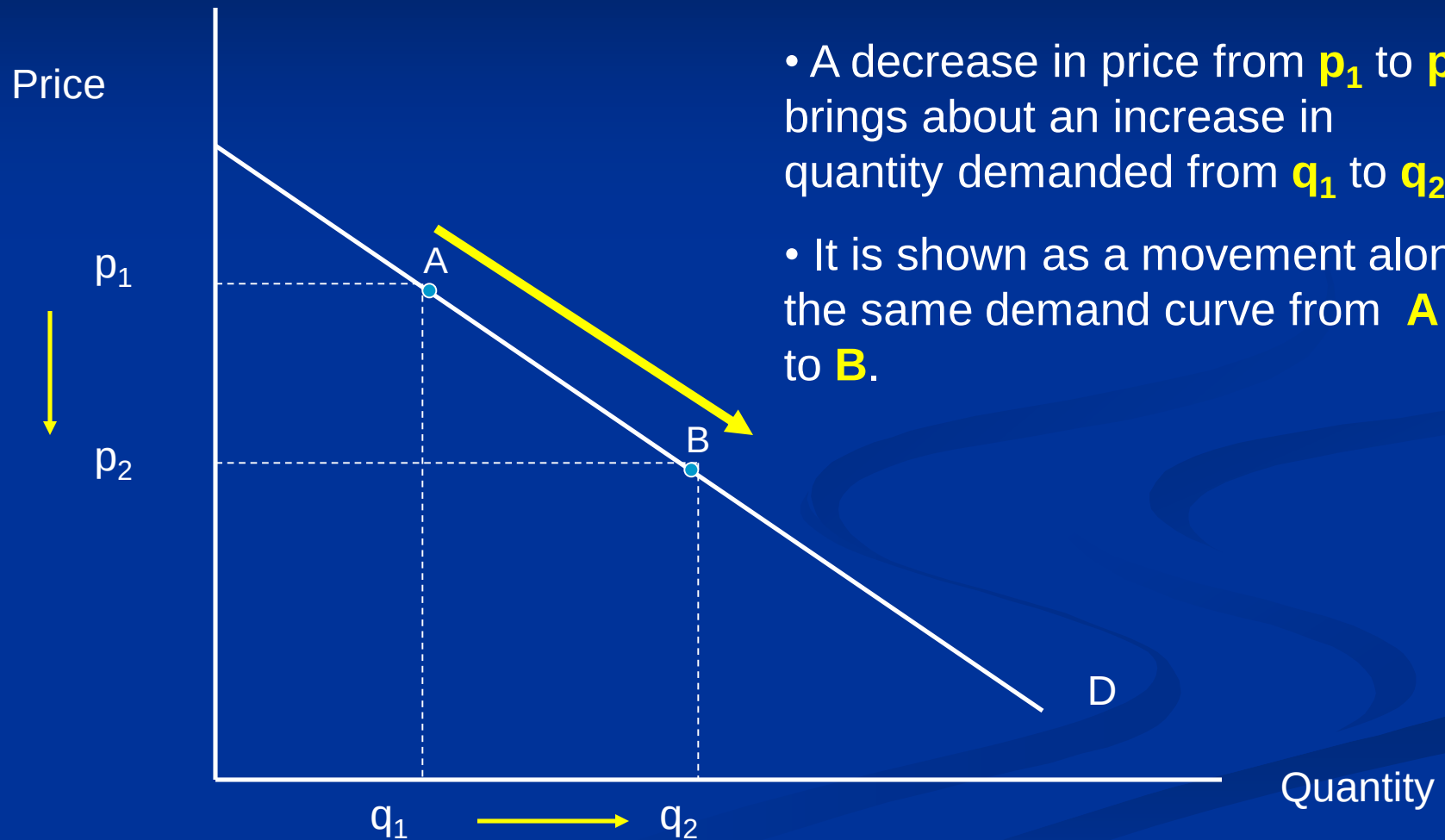
- Example: demand for housing and transportation increases with an **increase in population**.

- On the other hand, less consumers will cause the market demand to decrease, resulting in a shift to the left of the entire demand curve.

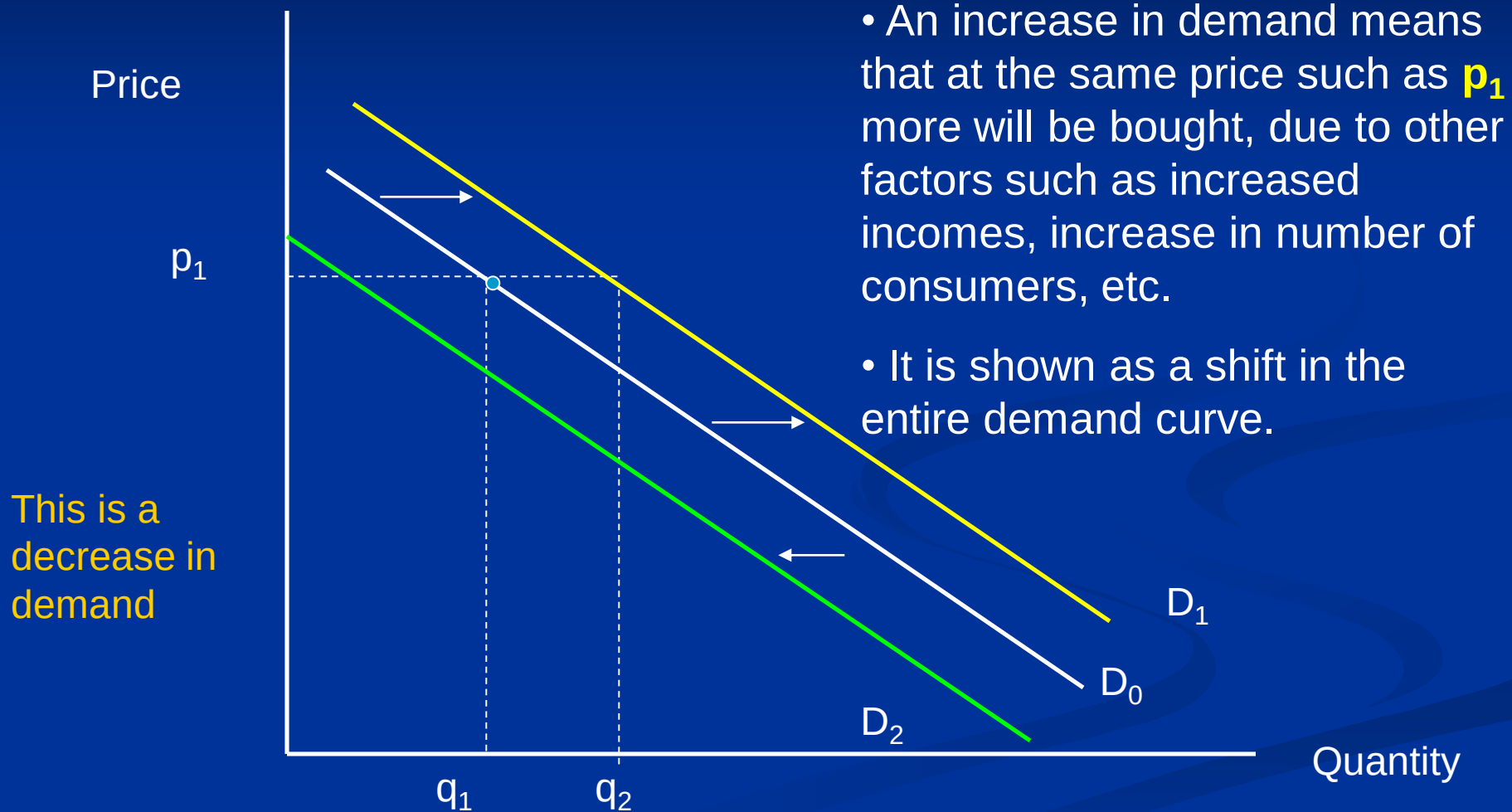
Change in Quantity Demanded vs. Change in Demand:

- **Change in quantity demanded** – is a movement along the same demand curve, due solely to a change in price, all other factors held constant.
- **Change in demand** – is a shift in the entire demand curve (either to the left or to the right) as a result of changes in other factors affecting demand.

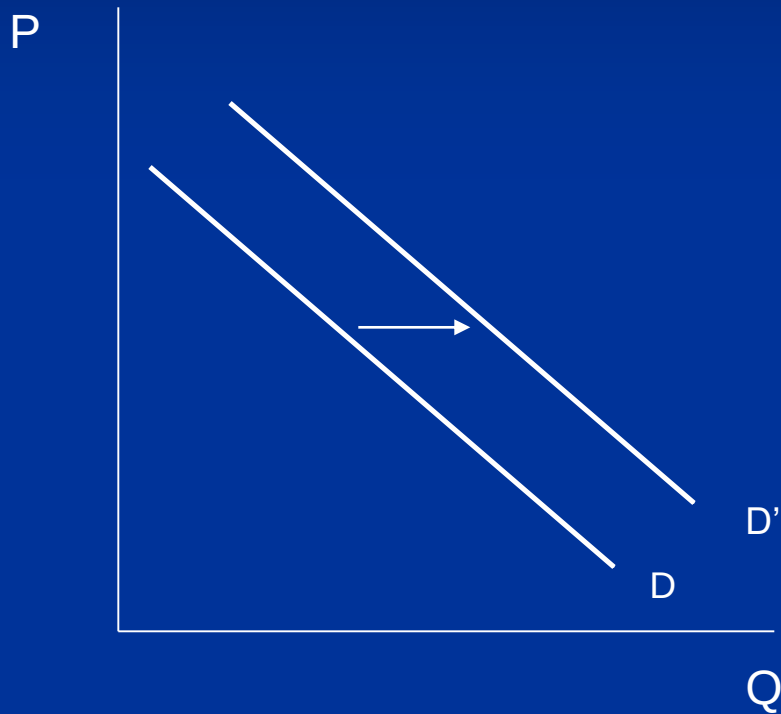
Change in Quantity Demanded:



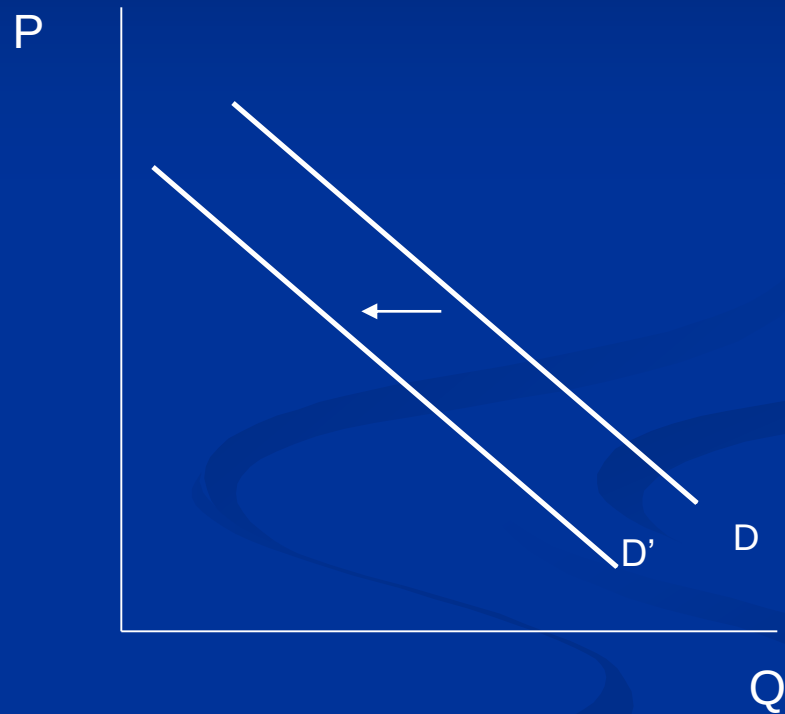
Change in Demand:



Change in Demand:



Increase in Demand



Decrease in Demand

The Concept of SUPPLY:

- **Supply** refers to the various quantities of a good or service that producers are willing to sell at alternative prices, *ceteris paribus*.
 - Obviously, firms are motivated to produce and sell more at higher prices.
 - Supply emphasizes the relationship between quantity sold of a commodity and its price. However, there are other factors that determine how much a producer would like to produce and sell.

The Law of Supply:

- It states that the quantity sold of a good or service is positively or directly related to its own price.
 - When the price increases, more of the good or service will be supplied.
 - When the price decreases, less of the commodity will be supplied.

Presentation of the Supply Relationship:

The relationship between **quantity supplied** and alternative **prices** may be presented in 3 ways:

- Supply schedule –in tabular form.
- Supply curve – in graphical form
- Supply function – in equation form

Supply Schedule:

Supply Schedule of Shoes

Price of a pair of shoes (in pounds)	Quantity supplied/year
0	0
50	1
100	2
150	3
200	4
250	5
300	6
350	7
400	8

Supply Curve:



Figure 2: Supply Curve. The positive slope of the supply curve depicts the direct relationship between price and quantity supplied.

Supply Function:

- Quantity supplied (Q_s) is expressed as a mathematical function of price (P). The supply function may thus be written as:

$$Q_s = c + dP$$

where

- c is the horizontal intercept of the equation or the quantity supplied when price is zero
 - d is the slope of the function.
- Example: $Q_s = 0 + 0.02P$

Factors Affecting Supply:

- There are other factors aside from price that affect the supply schedule. Some of the most important of these factors include:
 1. Resource prices.
 2. Prices of related goods in production.
 3. Technology.
 4. Expectations.
 5. Number of sellers.

(Cont.)

■ Resource Prices:

- When prices of inputs to production increase, the supply of the firm's product decreases.
- Decreases in resource prices, however, translate into an increase in supply. The entire supply curve shifts to the right.

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■ Prices of Related Goods in Production:

- Resources can be employed to produce several alternative goods and services.
- Examples from agriculture and industry:
 - a piece of farmland can be used to grow cotton, corn, or sugarcane. An increase in price of sugarcane may result in decreased supply of cotton and corn.
 - farmers can use their land and labor to produce ornamental flowers instead of vegetables. If vegetable prices decrease, the supply of ornamental flowers may increase.
 - An increase in the price of cement will increase supply of steel.

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■ Technology:

A change in production techniques can lower or raise production costs and affect supply.

- Improvements in technology shift the supply curve to the right.
 - A cost-saving invention will enable firms to produce and sell more goods than before at any given price.
 - New high yielding crop varieties will increase production on the same amount of land.

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■ Producer Expectations:

- When producers expect the price of their product to increase in the future, they may hoard their output for later sale, thus reducing supply in the present period. Thus the supply curve shifts to the left.
- If firms expect that the price of their product will fall in the near future, supply may increase in the current period as firms try to increase production as well as to dispose of their inventory.

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■ Number of Sellers:

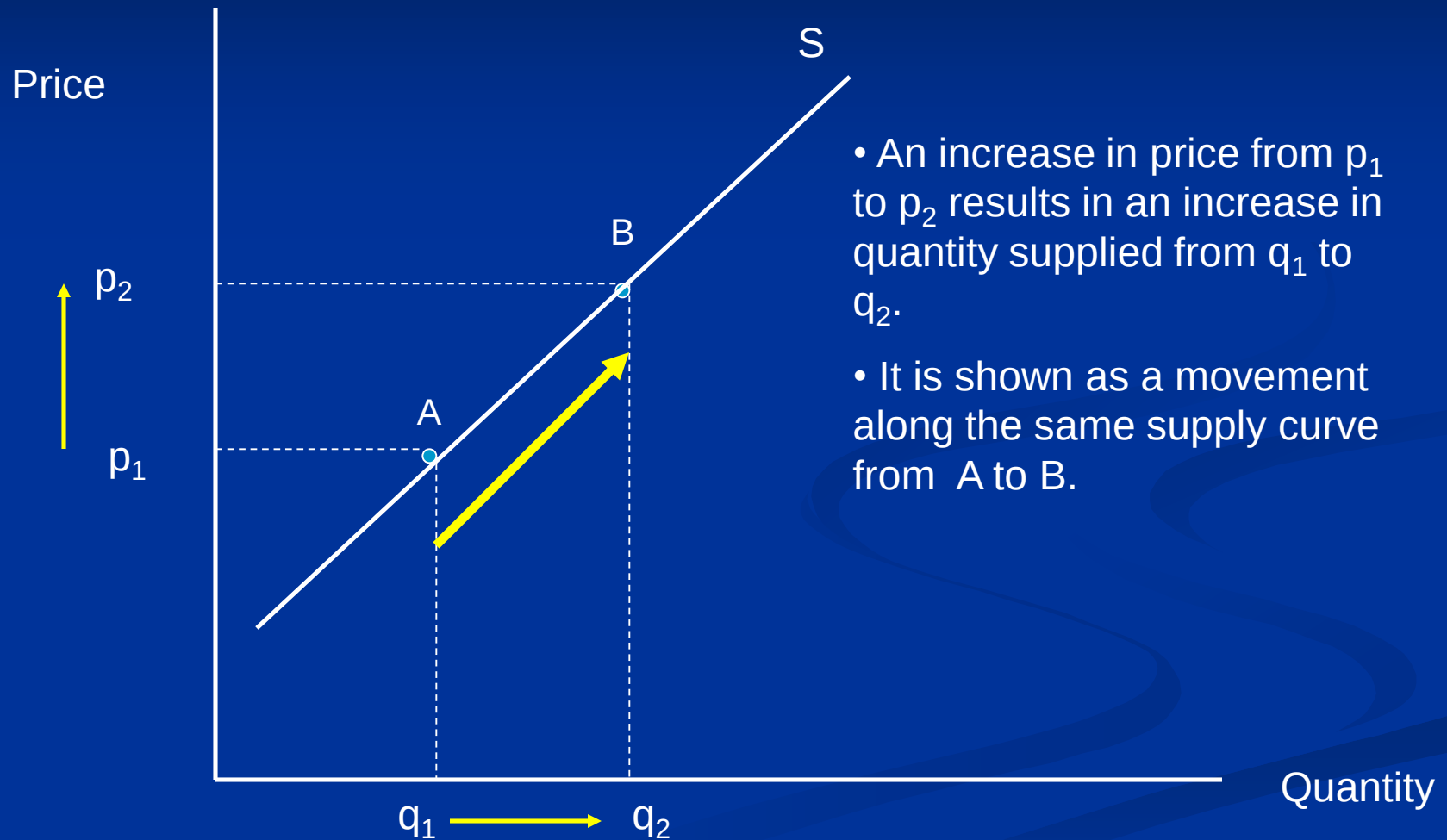
As the number of sellers increases, so will total supply.

- The market supply is the horizontal summation of the supply schedules of individual producers.
- As more firms enter the market, more will be offered for sale at each possible price, thus shifting the supply curve to the right.
- Similarly, the supply curve shifts to the left when firms exit the market.

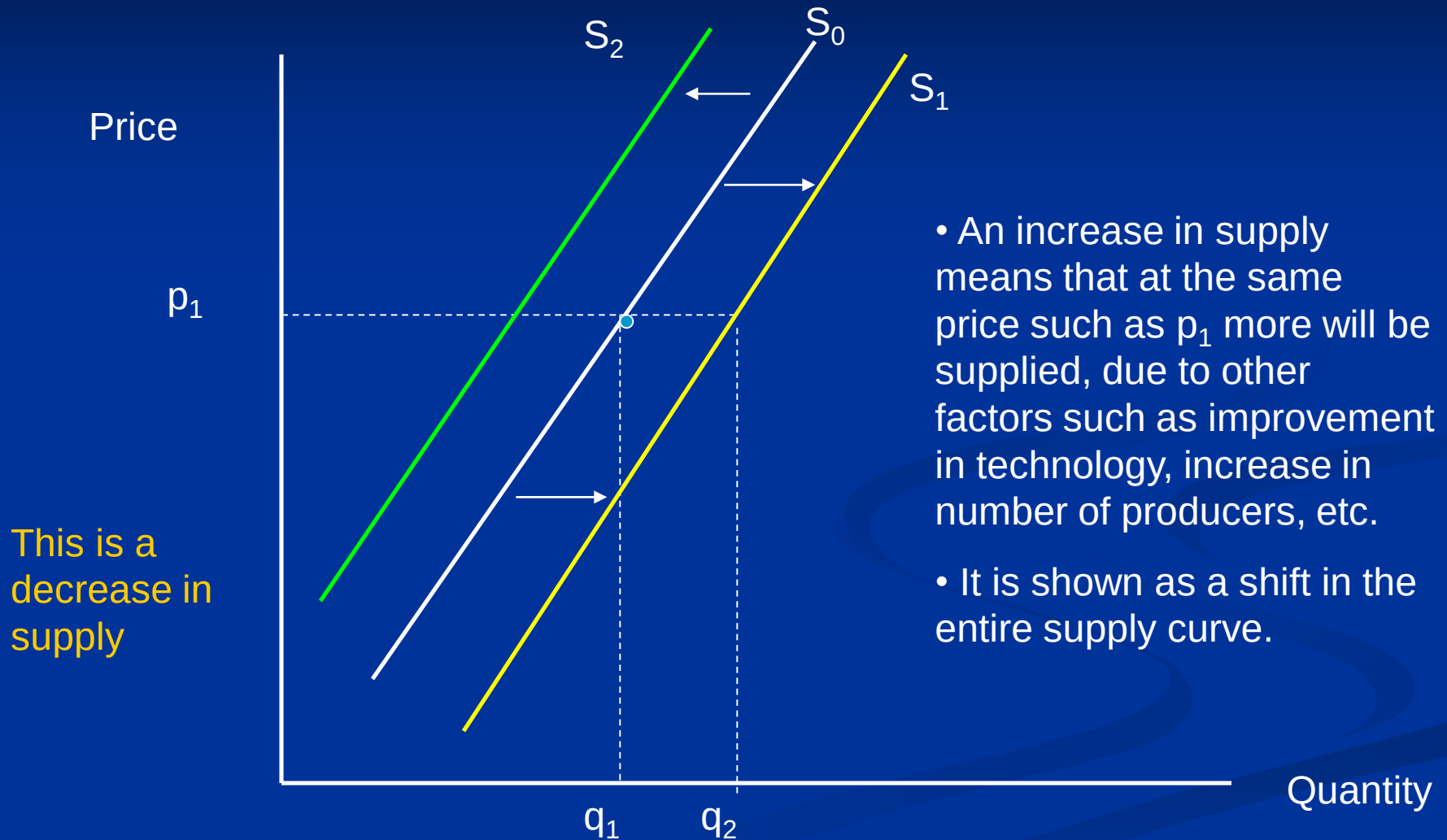
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- **Change in supply** – is a shift in the entire supply curve (either to the left or to the right) as a result of changes in other factors affecting supply.

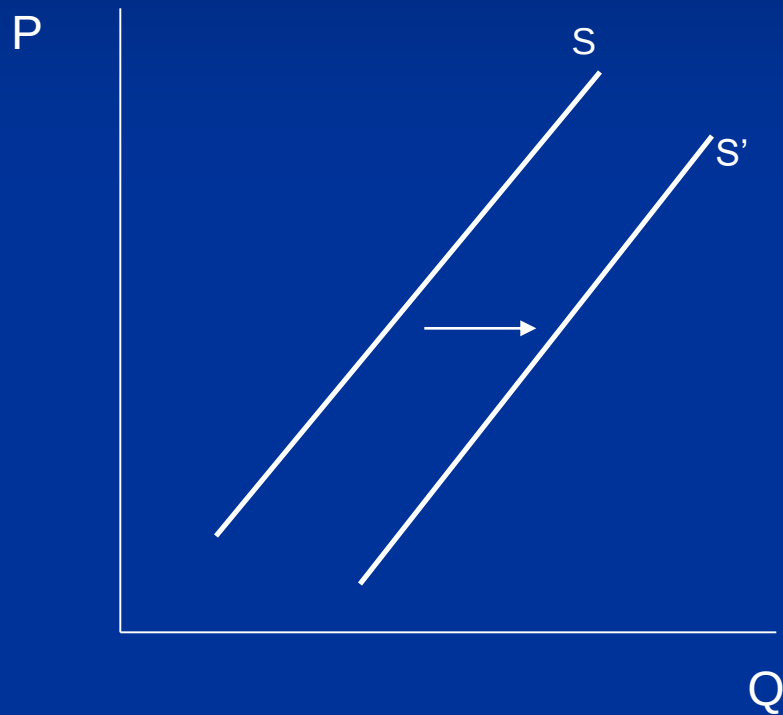
Change in Quantity Supplied:



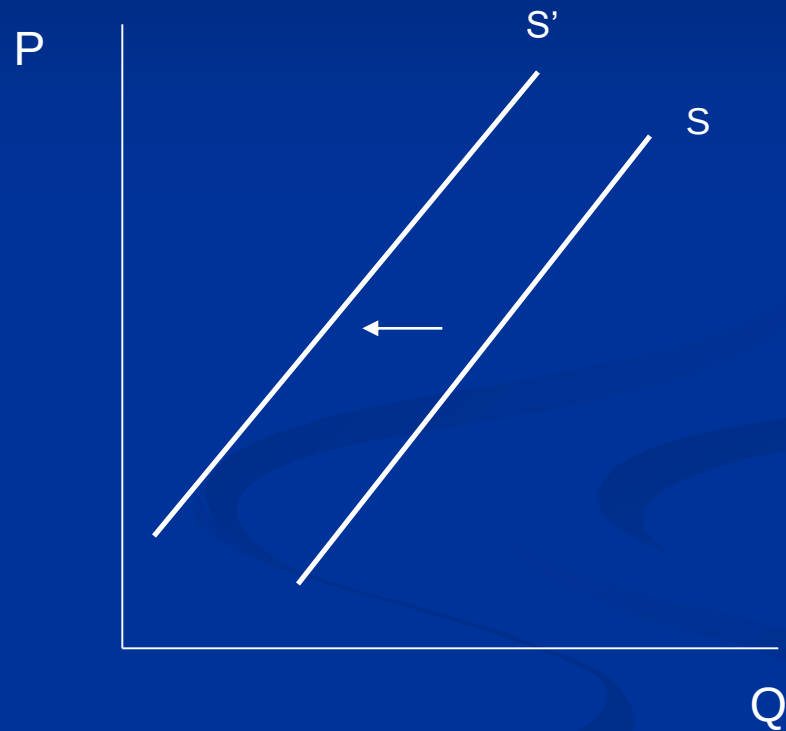
Change in Supply:



(Cont.)



Increase in Supply



Decrease in Supply

Market Equilibrium

- **Market equilibrium** is that state in which the quantity that firms want to supply equals the quantity that consumers want to buy.
 - The price that clears the market is called the ***equilibrium price*** and the quantity (sold and bought) is called the ***equilibrium quantity***.
 - The market is said to be "at rest" since the equilibrium price and equilibrium quantity will stay at those levels until either demand or supply changes.

(Cont.)

Market for Shoes

Price of a Pair of Shoes (in pounds)	Quantity Demanded per Year	Quantity Supplied per Year
0	8	0
50	7	1
100	6	2
150	5	3
200	4	4
250	3	5
300	2	6
350	1	7
400	0	8

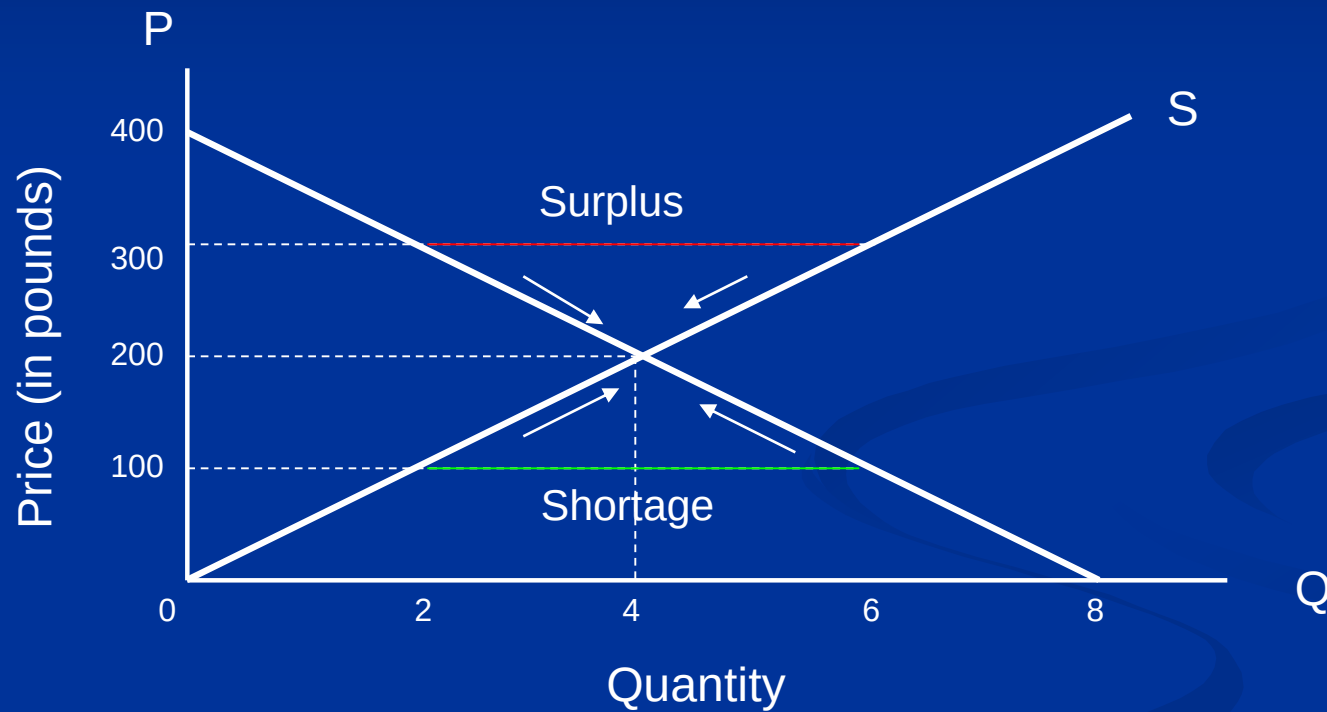
Equilibrium
Price=200

Equilibrium Quantity=4

(Cont.)

- At prices above the equilibrium price, quantity supplied is greater than quantity demanded, resulting in a temporary surplus.
 - In a surplus situation, producers will try to reduce price to entice consumers to buy more shoes. Actions by both producers and the public will wipe out the temporary surplus.
- At prices below the equilibrium price, consumers desire to buy more shoes than are available, creating a temporary shortage.
 - Consumers will try to outbid each other, thus pushing up the price. As price rises, firms increase their production while some consumers reduce their purchases.

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- Algebraic solution: equate the demand and supply equations ($Q_d = Q_s$).

$$Q_d = 8 - 0.02P$$

$$Q_s = 0 + 0.02P$$

- *Step by step solution:*

- $8 - 0.02P = 0 + 0.02P$

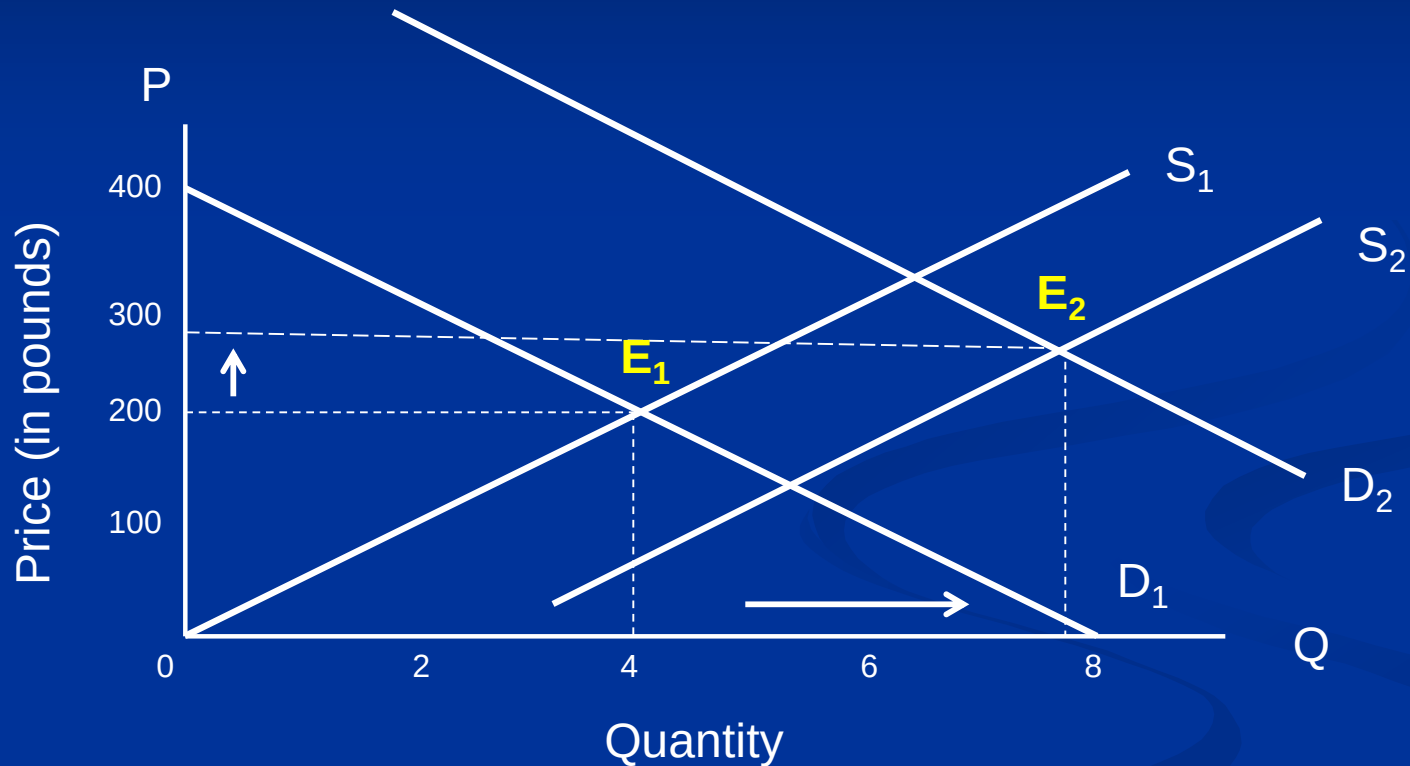
- $0.04P = 8$

- $P^* = 8/0.04 = 200$

- $Q_d = 8 - 0.02(200) = 8 - 4 = 4$

- $P^* = 200$ per unit, $Q^* = 4$ per year

Changes in Both Demand and Supply:



Changes in both demand and supply will move equilibrium to a new level, and the new equilibrium price and quantity will be determined by the direction and magnitude of the change in demand and supply.

Readings: Samuelson & Nordhaus, Ch. 3